

LECTURE 6

Monotone Convergence, the Squeeze Theorem, and the Upper and Lower Limits

MATH S4061 | Introduction to Modern Analysis I

Thursday, 11 June 2026

Of the small there is no smallest, but always a smaller; and of the large there is always a larger.

ANAXAGORAS, FRAGMENT B3 (5TH C. BCE)

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Overview. Picking up from completeness, this lecture assembles the basic machinery for limits of real sequences. Monotonicity plus boundedness already forces convergence; the comparison and squeeze principles let limits be read off from neighbours; and for sequences that oscillate, the upper and lower limits $\limsup$ and $\liminf$ record the band the sequence is eventually trapped in, coinciding exactly when the sequence converges. After recording four limits that recur throughout the subject, we open the theory of series: convergence through partial sums, the Cauchy criterion, and the first divergence test. The companion text is Rudin, Chapter 3, especially R 3.12–R 3.40.

6.1 Review of Completeness and Cauchy Sequences

Definition 6.1.1

Complete metric space

A metric space (X, d) is *complete* (R 3.12) if every Cauchy sequence in X converges to a point of X .

Example 6.1.2

Complete and incomplete spaces

- Compact metric spaces are complete (R 3.11).
- $\mathbb{R}^k$ is complete (R 3.11).
- $\mathbb{Q}$ and $(0, 1)$ are *not* complete (cf. R 3.12).

Remark 6.1.3

Testing convergence without the limit

In a *complete space* — in particular in $\mathbb{R}^k$ — it is enough to check whether a sequence is Cauchy; there is no need to know the limit in advance. In an *incomplete space* such as $\mathbb{Q}$ this fails: a Cauchy sequence may have no limit (6.1.2; cf. R 3.11–R 3.12).

6.2 Monotone sequences

Definition 6.2.1

Monotonic sequences

A sequence $\{s_n\}$ is *monotonic* in Rudin’s sense (R 3.13) as follows:

- *monotonic increasing* if $s_n \leq s_{n+1}$ for all n ;
- *monotonic decreasing* if $s_n \geq s_{n+1}$ for all n .

“Monotonic” means either increasing or decreasing.

Theorem 6.2.2

Monotone convergence

If $\{s_n\} \subseteq \mathbb{R}$ is monotonic and bounded, then $\{s_n\}$ converges (R 3.14).

Proof 6.2.2

Proof by a Supremum Argument of the Monotone Convergence Theorem.

- (1) Treat the increasing case (the decreasing case is analogous). Since $\{s_n\}$ is bounded above, the least-upper-bound property of $\mathbb{R}$ provides

$$s := \sup_n s_n.$$

- (2) Let $\varepsilon > 0$. Because $s - \varepsilon < s$, the number $s - \varepsilon$ is not an upper bound of $\{s_n\}$, so there is N_ε with

$$s_{N_\varepsilon} > s - \varepsilon.$$

(3) As the sequence is increasing and s is an upper bound, for every $n \geq N_\varepsilon$

$$s - \varepsilon < s_{N_\varepsilon} \leq s_n \leq s,$$

hence $|s_n - s| < \varepsilon$. Therefore $s_n \rightarrow s$. ■

[A5] cf. R 3.14; the decreasing case follows by applying the increasing case to $\{-s_n\}$.

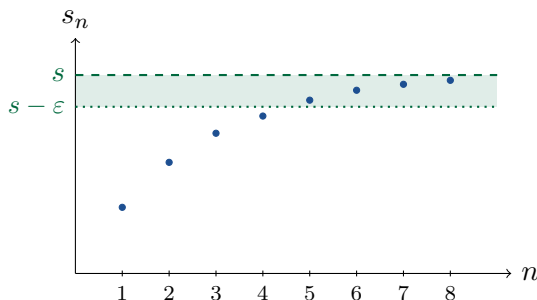


Figure 6.2.3. CLIMBING TO THE SUPREMUM. An increasing sequence bounded above closes in on $s = \sup_n s_n$: the terms rise but never cross the dashed bound, and from some N_ε on they all lie in the band $(s - \varepsilon, s]$.

6.3 Comparison and squeeze

Proposition 6.3.1

Order is preserved in the limit

Let $\{s_n\}$ and $\{t_n\}$ be convergent real sequences with $s_n \leq t_n$ for all n . Then

$$\lim_{n \rightarrow \infty} s_n \leq \lim_{n \rightarrow \infty} t_n.$$

This is the ordinary-limit form of Rudin's comparison theorem for upper and lower limits (cf. R 3.19).

Idea. Suppose instead that $s := \lim s_n$ and $t := \lim t_n$ satisfy $s > t$. Separate s and t by disjoint ε -balls, with $\varepsilon < \frac{1}{2}|s - t|$, so that $B_\varepsilon(t)$ and $B_\varepsilon(s)$ sit on opposite sides of the midpoint $\frac{s+t}{2}$.

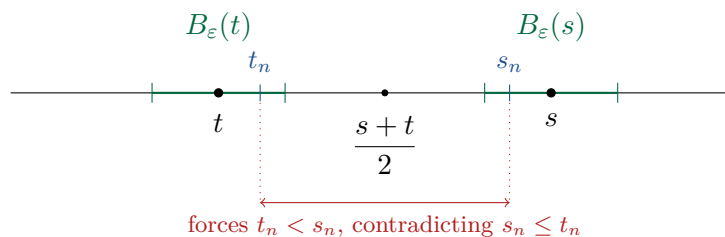


Figure 6.3.2. SEPARATING THE LIMITS. With $\varepsilon < \frac{1}{2}|s - t|$, the balls $B_\varepsilon(t)$ and $B_\varepsilon(s)$ fall on opposite sides of the midpoint $\frac{s+t}{2}$, which is impossible once $s_n \leq t_n$.

Proof 6.3.1

Proof by Contradiction that Order Passes to the Limit.

- (1) Write $s = \lim s_n$, $t = \lim t_n$, and suppose for contradiction $s > t$. Since $s - t > 0$, choose

$$0 < \varepsilon < \frac{1}{2}|s - t| = \frac{1}{2}(s - t).$$

- (2) By convergence there are N_1, N_2 with $|s_n - s| < \varepsilon$ for $n \geq N_1$ and $|t_n - t| < \varepsilon$ for $n \geq N_2$. With $N = \max\{N_1, N_2\}$, for every $n \geq N$

$$s_n > s - \varepsilon \quad \text{and} \quad t_n < t + \varepsilon.$$

- (3) The choice of ε puts these on opposite sides of the midpoint:

$$t + \varepsilon < t + \frac{1}{2}(s - t) = \frac{s + t}{2} = s - \frac{1}{2}(s - t) < s - \varepsilon.$$

- (4) Chaining the inequalities, for every $n \geq N$,

$$t_n < t + \varepsilon < \frac{s + t}{2} < s - \varepsilon < s_n,$$

so $t_n < s_n$, contradicting $s_n \leq t_n$. Hence $s \leq t$, i.e. $\lim s_n \leq \lim t_n$. ■

[A9]

Proposition 6.3.3

Squeeze theorem

Suppose $r_n \leq s_n \leq t_n$ for all n , where $r_n \rightarrow \ell$ and $t_n \rightarrow \ell$. Then $s_n \rightarrow \ell$. This is the comparison principle used in Rudin's special-sequence computations (cf. R 3.20).

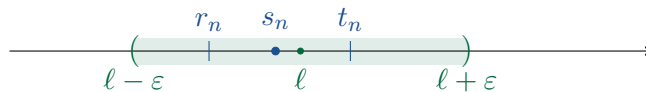


Figure 6.3.4. THE SQUEEZE. Once r_n and t_n lie in $(\ell - \varepsilon, \ell + \varepsilon)$, the trapped term s_n must lie there too.

Proof 6.3.3

Proof by an ε - N Estimate of the Squeeze Theorem.

- (1) Let $\varepsilon > 0$. Since $r_n \rightarrow \ell$ and $t_n \rightarrow \ell$, there is N with, for all $n \geq N$,
- $$|r_n - \ell| < \varepsilon \quad \text{and} \quad |t_n - \ell| < \varepsilon,$$
- so $r_n, t_n \in (\ell - \varepsilon, \ell + \varepsilon)$.
- (2) Using $r_n \leq s_n \leq t_n$, for every $n \geq N$
- $$\ell - \varepsilon < r_n \leq s_n \leq t_n < \ell + \varepsilon,$$
- hence $|s_n - \ell| < \varepsilon$. As $\varepsilon > 0$ was arbitrary, $s_n \rightarrow \ell$. ■
- [A2]

6.4 The upper and lower limits

Convergence $s_n \rightarrow \ell$ says that for every $\varepsilon > 0$ there is N with $s_n \in (\ell - \varepsilon, \ell + \varepsilon)$ for $n \geq N$. For a bounded sequence that does *not* converge, no single ℓ works. Instead there are two values $\ell^- \leq \ell^+$ such that the sequence is eventually trapped in any ε -enlargement of the band $[\ell^-, \ell^+]$:

$$\forall \varepsilon > 0 \quad \exists N \text{ s.t. } s_n \in (\ell^- - \varepsilon, \ell^+ + \varepsilon) \quad \text{for } n \geq N.$$

We write $\ell^- = \liminf s_n$ and $\ell^+ = \limsup s_n$.

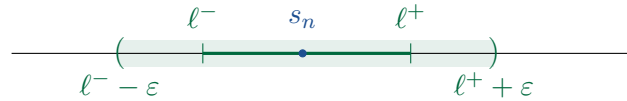


Figure 6.4.1. THE TRAPPING BAND. For a bounded sequence, every ε -enlargement of $[\ell^-, \ell^+]$ eventually contains all the terms.

The two coincide exactly when the sequence converges:

$$s_n \rightarrow \ell \iff \liminf s_n = \limsup s_n = \ell.$$

Example 6.4.2

An oscillating sequence

For $s_n = (-1)^n \frac{n}{n-1}$ (with $n \geq 2$), the even terms approach $+1$ from above and the odd terms approach -1 from below. Hence $\liminf s_n = -1$ and $\limsup s_n = +1$, and the sequence does not converge, since these differ (cf. R 3.18).

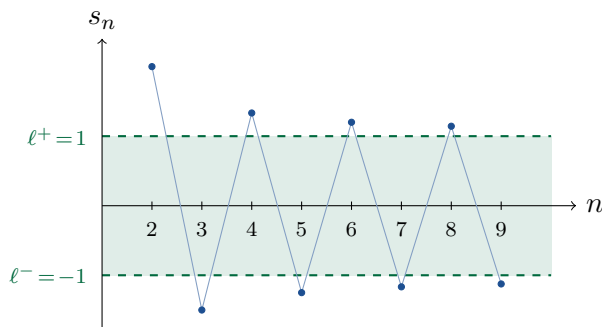


Figure 6.4.3. TWO LIMIT POINTS. The oscillating sequence $s_n = (-1)^n \frac{n}{n-1}$ has no limit: the even terms fall toward $\ell^+ = \limsup s_n = 1$ from above and the odd terms rise toward $\ell^- = \liminf s_n = -1$ from below. Every ε -enlargement of the shaded band $[\ell^-, \ell^+]$ eventually traps the whole tail.

Definition 6.4.4

Divergence to $\pm\infty$

For a real sequence $\{t_n\}$ (R 3.15),

- $t_n \rightarrow +\infty$ means: for every $M > 0$ there is N with $t_n > M$ for all $n \geq N$;
- $t_n \rightarrow -\infty$ means: for every $M > 0$ there is N with $t_n < -M$ for all $n \geq N$.

Such sequences diverge, but in a specific, controlled way.

Definition 6.4.5

Limit inferior and limit superior

The *upper* and *lower limits* of a real sequence $\{s_n\}$ are defined by the tail extrema

$$\limsup s_n = \lim_{n \rightarrow \infty} \left(\sup_{k \geq n} s_k \right), \quad \liminf s_n = \lim_{n \rightarrow \infty} \left(\inf_{k \geq n} s_k \right).$$

The tails are nested, $\{s_k : k \geq n+1\} \subseteq \{s_k : k \geq n\}$, so $\sup_{k \geq n} s_k$ is non-increasing in n (and $\inf_{k \geq n} s_k$ non-decreasing); hence both limits always exist in the extended reals $[-\infty, +\infty]$. Two characterizations follow.

(1) Subsequential limits. If $\{s_n\}$ is bounded and E is its set of subsequential limits, then

$$\liminf s_n = \min E, \quad \limsup s_n = \max E.$$

(2) Eventual trapping (finite $\liminf, \limsup$). For every $\varepsilon > 0$ there is N_ε with

$$s_n \in (\liminf s_n - \varepsilon, \limsup s_n + \varepsilon) \quad \text{for } n \geq N_\varepsilon,$$

and $[\liminf s_n, \limsup s_n]$ is the smallest closed interval with this property.

This tail-extrema presentation is equivalent to Rudin's subsequential-limit definition (R 3.16) and characterization (R 3.17).

Remark 6.4.6

The extreme limits are attained

For a *bounded* sequence, E is closed, bounded, and nonempty, so it contains its own extrema: $\inf E \in E$ and $\sup E \in E$. Hence $\liminf s_n$ and $\limsup s_n$ are themselves subsequential limits of $\{s_n\}$. (For an unbounded sequence this holds in the extended reals, with $\pm\infty$ admitted as subsequential limits.) Compare R 3.17.

Remark 6.4.7

Existence

If $\{s_n\}$ is bounded, then E is bounded and nonempty, so $\liminf s_n$ and $\limsup s_n$ exist as finite reals. If $\{s_n\}$ is unbounded, then E may include $+\infty$ and/or $-\infty$, and $\liminf s_n, \limsup s_n$ may equal $\pm\infty$ (cf. R 3.16–R 3.17).

6.5 Some standard limits

Proposition 6.5.1

Four standard limits

- (a) If $|x| < 1$, then $x^n \rightarrow 0$.
- (b) If $p > 0$, then $\frac{1}{n^p} \rightarrow 0$.
- (c) If $p > 1$, then $\sqrt[p]{p} \rightarrow 1$.
- (d) $\sqrt[p]{n} \rightarrow 1$.

These are the recurring special limits collected in R 3.20(a)–(c),(e).

Proof 6.5.1.a

Proof by the Binomial Inequality of $x^n \rightarrow 0$.

- (1) If $x = 0$ then $x^n = 0 \rightarrow 0$, so assume $0 < |x| < 1$. Then $1/|x| > 1$, so $|x| = \frac{1}{1+p}$ for a unique $p > 0$ (the map $p \mapsto \frac{1}{1+p}$ is a bijection $(0, \infty) \rightarrow (0, 1)$).

- (2) By the binomial theorem, keeping the nonnegative $k = 0, 1$ terms,

$$(1+p)^n = \sum_{k=0}^n \binom{n}{k} p^k \geq 1 + np.$$

- (3) Hence

$$|x|^n = \frac{1}{(1+p)^n} \leq \frac{1}{1+np} \xrightarrow{n \rightarrow \infty} 0,$$

and since $|x^n| = |x|^n \rightarrow 0$, we conclude $x^n \rightarrow 0$. ■

[A2]

Proof 6.5.1.b

Proof by an ε - N Estimate of $1/n^p \rightarrow 0$.

(1) Let $\varepsilon > 0$ and choose N with $N > (1/\varepsilon)^{1/p}$, equivalently $\frac{1}{N^p} < \varepsilon$.

(2) Then for every $n \geq N$,

$$\frac{1}{n^p} \leq \frac{1}{N^p} < \varepsilon,$$

$$\text{so } \frac{1}{n^p} \rightarrow 0.$$

[A2]

Proof 6.5.1.c

Proof by the Squeeze Theorem of $\sqrt[p]{p} \rightarrow 1$.

(1) Assume $p > 1$ and set $x_n = \sqrt[p]{p} - 1 \geq 0$. By the binomial inequality from (a),

$$1 + nx_n \leq (1 + x_n)^n = p,$$

$$\text{so } 0 \leq x_n \leq \frac{p-1}{n}.$$

(2) Since $\frac{p-1}{n} \rightarrow 0$, the squeeze theorem (6.3.3) gives $x_n \rightarrow 0$, i.e. $\sqrt[p]{p} = 1 + x_n \rightarrow 1$.

[A2] uses 6.3.3.

Proof 6.5.1.d

Proof by the Squeeze Theorem of $\sqrt[n]{n} \rightarrow 1$.

(1) Set $x_n = \sqrt[n]{n} - 1 \geq 0$, so $n = (1 + x_n)^n$. The linear term is too weak, so keep the $k = 2$ term:

$$n = (1 + x_n)^n \geq \binom{n}{2} x_n^2 = \frac{n(n-1)}{2} x_n^2 \quad (n \geq 2).$$

(2) Rearranging, $x_n^2 \leq \frac{2}{n-1}$, hence $0 \leq x_n \leq \sqrt{\frac{2}{n-1}}$.

(3) Since $\sqrt{\frac{2}{n-1}} \rightarrow 0$, the squeeze theorem (6.3.3) gives $x_n \rightarrow 0$, i.e. $\sqrt[n]{n} = 1 + x_n \rightarrow 1$.

[A2] keep the $k = 2$ binomial term; uses 6.3.3.

Remark 6.5.2

The case $0 < p < 1$ in (c)

For $0 < p < 1$, apply part (c) to $\sigma = 1/p > 1$: since $\sqrt[p]{\sigma} \rightarrow 1$, also $\sqrt[p]{p} = 1/\sqrt[p]{\sigma} \rightarrow 1$ (and $\sqrt[p]{1} = 1$).

Hence $\sqrt[p]{p} \rightarrow 1$ for every $p > 0$; the full $p > 0$ statement is R 3.20(b).

6.6 Introduction to series

(Not covered on the midterm.)

As a field, $\mathbb{R}$ hands us directly only the polynomials $1, x, x^2, \dots$ and, as their quotients, the rational functions $P(x)/Q(x)$. What about e^x , $\sin x$, $\cos x$? These are reached not by field operations but by infinite sums — for instance

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}.$$

To make sense of such a sum we must first say what an infinite sum *is*.

Definition 6.6.1

Series and partial sums

Given a sequence $\{a_n\}$ in $\mathbb{R}$, its *k-th partial sum* is

$$s_k = a_1 + \cdots + a_k = \sum_{i=1}^k a_i \quad (s_0 = 0, \text{ the empty sum}).$$

If $s_k \rightarrow s$, the *series* $\sum a_n$ *converges*, and we write

$$\sum_{i=1}^{\infty} a_i = \lim_{k \rightarrow \infty} s_k = s.$$

A series may instead be indexed from any starting point—often $n = 0$, as for $\sum c_n x^n$ —with the partial sums adjusted accordingly. This is Rudin's series convention (R 3.21) with the real-valued specialization used in lecture.

Theorem 6.6.2

Cauchy criterion for series

A series $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is N such that

$$\left| \sum_{k=m}^n a_k \right| < \varepsilon \quad \text{for all } n \geq m \geq N.$$

This is the Cauchy criterion for series (R 3.22).

Proof 6.6.2

Proof by Completeness of $\mathbb{R}$ of the Cauchy Criterion.

- | | |
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| <div style="border-left: 1px solid black; height: 100px; margin-left: 5px;"></div> | <p>(1) For $n \geq m \geq 1$ one has $s_n - s_{m-1} = \sum_{k=m}^n a_k$ (with $s_0 = 0$), so the displayed condition says precisely that the partial sums $\{s_k\}$ form a Cauchy sequence.</p> <p>(2) Since $\mathbb{R}$ is complete (6.1.1), $\{s_k\}$ converges if and only if it is Cauchy. Hence $\sum a_n$ converges if and only if the criterion holds.</p> |
|--|---|

■

[Direct] cf. R 3.22; the partial sums converge iff they are Cauchy, and $\mathbb{R}$ is complete.

Corollary 6.6.3

Vanishing of the terms

If $\sum a_n$ converges, then $a_n \rightarrow 0$ (R 3.23).

Proof 6.6.3

Proof by the Cauchy Criterion with $m = n$.

- (1) Take $m = n$ in the criterion (6.6.2): for every $\varepsilon > 0$ there is N with

$$|a_n| = \left| \sum_{k=n}^n a_k \right| < \varepsilon \quad \text{for all } n \geq N.$$

- (2) Therefore $a_n \rightarrow 0$. ■

[Direct] cf. R 3.23.

So $a_n \rightarrow 0$ is *necessary* for convergence. Is it *sufficient*? No.

Proposition 6.6.4

The harmonic series diverges

Although $\frac{1}{n} \rightarrow 0$, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. This is the $p = 1$ case of Rudin's p -series theorem (R 3.28).

Proof 6.6.4

Proof by Dyadic Grouping of the Divergence of the Harmonic Series.

- (1) Round each term down to the next power of $\frac{1}{2}$: replace $\frac{1}{3}$ by $\frac{1}{4}$; then $\frac{1}{5}, \frac{1}{6}, \frac{1}{7}$ by $\frac{1}{8}$; and so on. If s'_k denotes the resulting partial sums, then $s_k \geq s'_k$ for every k .

- (2) The rounded series groups into dyadic blocks of equal weight:

$$1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots = 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots,$$

each block summing to $\frac{1}{2}$; hence $s'_{2^m} = 1 + \frac{m}{2}$.

- (3) Therefore $s_{2^m} \geq s'_{2^m} = 1 + \frac{m}{2} \rightarrow \infty$, so the partial sums $\{s_n\}$ have a subsequence diverging to $+\infty$ and cannot converge. Thus $\sum \frac{1}{n}$ diverges. ■

[Direct] Cauchy condensation; cf. R 3.27–R 3.28.

Proposition 6.6.5

The p -series

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$. Rudin proves the full convergence/divergence dichotomy in R 3.28.

Proof 6.6.5

Proof of the Convergence of the p -Series.

$\leftrightarrow$ Stated in lecture as a fact, for use as a comparison series; cf. R 3.28.

Proposition 6.6.6

A nonnegative series converges when bounded

A series $\sum a_n$ of nonnegative terms ($a_n \geq 0$) converges if and only if its partial sums are bounded (R 3.24).

Proof 6.6.6

Proof by Monotone Convergence for a Series of Nonnegative Terms.

- (1) Since $a_{n+1} \geq 0$, the partial sums satisfy $s_{n+1} = s_n + a_{n+1} \geq s_n$, so $\{s_n\}$ is increasing.
- (2) If the partial sums are bounded, then $\{s_n\}$ is increasing and bounded, hence converges by the monotone convergence theorem (6.2.2); that is, $\sum a_n$ converges. Conversely, a convergent sequence is bounded.

■

[A5] cf. R 3.24; the partial sums increase; apply 6.2.2.

Proposition 6.6.7

The geometric series

For $|x| < 1$,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

This is the convergent half of Rudin's geometric-series theorem (R 3.26).

Proof 6.6.7

Proof by the Partial-Sum Formula of the Geometric Series.

- (1) For $x \neq 1$ the partial sums have the closed form

$$s_k = 1 + x + \cdots + x^k = \frac{1 - x^{k+1}}{1 - x}.$$

- (2) Since $|x| < 1$ gives $|x|^{k+1} \rightarrow 0$, we have $x^{k+1} \rightarrow 0$ and hence

$$s_k \xrightarrow{k \rightarrow \infty} \frac{1}{1-x},$$

$$\text{so } \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

■

[Direct] cf. R 3.26; uses $|x|^k \rightarrow 0$ for $|x| < 1$ (6.5.1).

Theorem 6.6.8

Comparison test for series

- (a) If $|a_n| \leq c_n$ for all $n \geq N_0$ and $\sum c_n$ converges, then $\sum a_n$ converges.
 (b) If $a_n \geq d_n \geq 0$ for all n and $\sum d_n$ diverges, then $\sum a_n$ diverges.

This is Rudin's comparison test (R 3.25).

Proof 6.6.8

Proof by the Cauchy Criterion of the Comparison Test.

- (1) For (a), let $\varepsilon > 0$. Since $\sum c_n$ converges, the Cauchy criterion (6.6.2) gives some $N \geq N_0$ with $\sum_{k=m}^n c_k < \varepsilon$ for all $n \geq m \geq N$. Then for such n, m ,

$$\left| \sum_{k=m}^n a_k \right| \leq \sum_{k=m}^n |a_k| \leq \sum_{k=m}^n c_k < \varepsilon,$$

so $\sum a_n$ meets the Cauchy criterion and therefore converges.

- (2) For (b), suppose instead $\sum a_n$ converged. Since $0 \leq d_n \leq a_n = |a_n|$, part (a) (with $c_n = a_n$) would force $\sum d_n$ to converge — contradicting that it diverges. Hence $\sum a_n$ diverges. ■

[A4] cf. R 3.25; bound the tail of $\sum a_n$ by that of $\sum c_n$.

The natural series to compare against is the geometric one. Comparing $\sum a_n$ to a geometric $\sum b_n$: if the ratio b_{n+1}/b_n is constant we are led to the *ratio test*, and if the root $\sqrt[n]{b_n}$ is constant, to the *root test*.

Theorem 6.6.9

Root test

Given $\sum a_n$, set $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Then

- if $\alpha < 1$, $\sum a_n$ converges;
- if $\alpha > 1$, $\sum a_n$ diverges;
- if $\alpha = 1$, the test is inconclusive.

This is Rudin's root test (R 3.33).

Proof 6.6.9

Proof by Comparison with a Geometric Series of the Root Test.

- (1) If $\alpha < 1$, choose β with $\alpha < \beta < 1$. Then $\sqrt[n]{|a_n|} < \beta$ for all large n , so $|a_n| < \beta^n$; as $\sum \beta^n$ converges, the comparison test (6.6.8) gives convergence.
 (2) If $\alpha > 1$, then $\sqrt[n]{|a_n|} > 1$ for infinitely many n , so $|a_n| > 1$ infinitely often and $a_n \not\rightarrow 0$; by 6.6.3 the series diverges. For $\alpha = 1$ the test decides nothing: both $\sum \frac{1}{n}$ (divergent) and $\sum \frac{1}{n^2}$ (convergent) have $\alpha = 1$. ■

[A1] cf. R 3.33; uses the comparison test (6.6.8).

Theorem 6.6.10**Ratio test**

Suppose $a_n \neq 0$ for all large n . Then $\sum a_n$ converges if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, and diverges if there is N with $\left| \frac{a_{n+1}}{a_n} \right| \geq 1$ for all $n \geq N$. This is Rudin's ratio test (R 3.34).

Proof 6.6.10

Proof by Comparison with a Geometric Series of the Ratio Test.

- | | |
|---|---|
| <div style="border-left: 1px solid black; height: 100px; margin-left: 10px;"></div> | <p>(1) If $\limsup a_{n+1}/a_n < 1$, choose β in between; then $a_{n+1}/a_n < \beta$ for $n \geq N$, so $a_n \leq a_N \beta^{n-N}$ for $n \geq N$. Comparison with the geometric series (6.6.8) gives convergence.</p> <p>(2) If $a_{n+1}/a_n \geq 1$ for all $n \geq N$, then $a_n \geq a_N > 0$ for $n \geq N$, so $a_n \not\rightarrow 0$ and the series diverges (6.6.3).</p> |
|---|---|

■

[A1] cf. R 3.34; uses the comparison test (6.6.8).

Definition 6.6.11**Power series and radius of convergence**

A *power series* is a function defined by

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots.$$

Fix $x \neq 0$ and apply the root test (6.6.9)¹ to $\sum c_n x^n$. With $\alpha := \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|}$,

$$\beta := \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n x^n|} = |x| \alpha,$$

so the series converges when $\beta = |x| \alpha < 1$ and diverges when $|x| \alpha > 1$. Hence it converges for $|x| < R$, where

$$R := \frac{1}{\alpha} \quad (R := +\infty \text{ if } \alpha = 0, \quad R := 0 \text{ if } \alpha = +\infty).$$

R is the *radius of convergence*: the series converges on $(-R, R)$ and diverges for $|x| > R$; at $x = 0$ it is just $f(0) = c_0$, and the endpoints $|x| = R$ require separate analysis. This is Rudin's power-series setup and radius formula (R 3.38–R 3.39), restricted here to real x .

Example 6.6.12*The exponential series*

¹The root test is used here because it is strictly more general than the ratio test; for the precise relationship see the appendix, §6.A.1.

For $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ the ratios have a limit, so they compute α (the valid case for the ratio shortcut):

$$\lim_{n \rightarrow \infty} \frac{1/(n+1)!}{1/n!} = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0 = \alpha,$$

so $R = 1/\alpha = +\infty$: the series converges for every x . This function is $f(x) = e^x$. Compare Rudin's examples following the radius-of-convergence theorem (R 3.40).

Appendix

Supplementary material—additional proofs, context, and other treatments; not part of the lecture proper.

Supplement 6.A.1

The root test subsumes the ratio test

For any sequence $\{c_n\}$ with $c_n \neq 0$,

$$\liminf_n \left| \frac{c_{n+1}}{c_n} \right| \leq \liminf_n \sqrt[n]{|c_n|} \leq \limsup_n \sqrt[n]{|c_n|} \leq \limsup_n \left| \frac{c_{n+1}}{c_n} \right|$$

(R 3.37). Hence whenever the ratio test settles convergence the root test settles it too, but not conversely; and if $\lim_n |c_{n+1}/c_n| = L$ exists then the root quantity tends to L as well, so the ratio test gives the same radius $R = 1/L$ (the converse fails — the radii can agree without the ratio limit existing). This is why 6.6.11 fixes R through the root quantity $\alpha = \limsup_n \sqrt[n]{|c_n|}$.

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